

Georges Parfait Djimefo

Kapen | Computer Scientist | AI Researcher

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Profile

Computer scientist with expertise in **AI**, **Machine Learning**, **Multi-Agent Reinforcement Learning**, and **6G network systems**. PhD expected Summer 2026. Strong background in software engineering, data science, cloud computing, and applied statistics. Proven ability to design novel AI architectures, publish original research, teach at the university level, and deliver production-grade ML pipelines. Innovation, teamwork, leadership, and results-oriented.

Technical Skills

Languages: Python, Java, C++, PHP, SQL, R, JavaScript, TypeScript

LLM / Agents: LangChain, LangGraph, LangSmith, MCP, RAG, Claude (Haiku, Opus), KIRO

AI / ML: TensorFlow, PyTorch, Scikit-Learn, MARL (QMIX, QTRAN), GNN, DQN, NLP, Computer Vision, LLMs

Data: PySpark, Databricks, Snowflake, Delta Lake, Relational Databases, ETL/ELT

Cloud: AWS (SageMaker, Bedrock AgentCore, CloudWatch), OpenStack, Kubernetes, Docker

DevOps: Jenkins, CI/CD, Git, GitHub Actions, GitLab

Web / Mobile: React, Angular, Node.js, D3.js, Plotly, Dash, Power BI, Tableau, Mobile Dev.

Methods: Agile/Scrum, TDD, Design Patterns, Statistical Modeling, Big Data, Data Visualization

Publications

[1] *Energy-Efficient and Near Real-Time Routing Algorithm for Disaster Monitoring in Wireless Sensor Networks*

G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal

(Submitted)

- Designed **GAPF**, a novel **meta-learning** framework fusing MARL policies (QMIX & QTRAN) via a **GCN encoder** with only **1 473 parameters**.
- Achieved $\geq 98\%$ packet delivery, $2\text{--}3.5\times$ less energy, $13\times$ less memory than baselines.

[2] *Carbon-Aware Autonomous Data Reduction and Self-Healing Routing for 6G-Integrated Disaster Sensor Networks*

G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal

(Submitted)

- Proposed **CALASH**, the **first protocol** jointly optimizing Lifecycle Carbon Intensity via a hybrid **Lyapunov-DQN** engine, **MAPE-K** self-healing, and ISO 14040 lifecycle analysis over 6G.
- Achieved 60% longer network lifetime and 33% lower LCI than baselines.

[3] *Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Networks*

G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal

(Submitted)

- Designed **CASTER-ZT**, the **first framework** combining **GNN-based decision making** with formally bounded **zero-trust** admissibility (10 proven properties).
- Achieved 74–98% rogue detection, zero false positives, 3 ms latency.

Professional Experience

Polytechnique Montréal

Lecturer

Montréal

2025–2026

- Teach the Python procedural programming course.

Intact Financial Corporation

AI Developer I

Montréal

2022–2024

- Ingest and process large-scale data.
- Monitor pipelines and develop automated testing solutions.
- Stack:* PySpark, SQL, Databricks, Snowflake, AWS, Shell, Git.

Polytechnique Montréal

Lecturer

Montréal

2022–2023

- Teach the Python procedural programming course.

Université de Montréal

Scientific Assistant — MOOC AI

Montréal

2021–2022

- Design courses and practical work in supervised learning (Python).

Ministry of Tourism and Recreation <i>Contractual Statistical Economist Engineer</i> ○ Analyze data to support policy decisions.	Cameroon 2012–2014
Protestant University of Central Africa <i>Associate Professor of Statistics & Probability</i> ○ Teach statistics and probability courses.	Cameroon 2010–2012
ISSEA (Sub-regional Institute of Statistics and Applied Economics) <i>Part-time Lecturer in Data Analysis</i> ○ Teach the Data Analysis course (R).	Cameroon 2009–2010

Internships & Selected Projects

Ericsson <i>Machine Learning Intern</i> ○ Build an LLM model and a RAG Agentic AI Agent. ○ Develop data pipelines and ML models; research and implement ML algorithms. ○ Test and optimize data & ML pipelines. ○ Collaborate with teams for maintainable, scalable code. ○ Ensure quality via unit/integration tests; continuous analysis and requirement handling. ○ Deploy on AWS SageMaker; monitor pipelines. ○ <i>Stack:</i> Python, AWS, SQL, GitLab, KIRO, LangChain, LangGraph, LangSmith, SageMaker, Bedrock AgentCore, MCP, Claude, CI/CD, CloudWatch.	Montréal 2026
Intact Financial Corporation <i>Data Engineering Intern</i> ○ Manipulate data (PySpark, SQL, Databricks, Snowflake, AWS).	Montréal 2022
Polytechnique Montréal <i>Undergraduate Research Initiation Intern</i> ○ “Modeling of Voice Recognition in Android systems to secure bank payments.” ○ Model with Deep Learning (Python); implement and deploy on Android mobile (Java).	Montréal 2020–2021
Polytechnique Montréal <i>Entrepreneur / Research Initiation Intern</i> ○ “Face Recognition modeling integrating racial and gender bias.” ○ Model with Deep Learning (Python).	Montréal 2020
Polytechnique Montréal <i>Integrator Project — IATA</i> ○ “Flight Data eXchange — Develop an initial and contextual flight data assessment tool.” ○ Contribute to software implementation (Dash, Python, React, Electron, Axios).	Montréal 2021
Polytechnique Montréal <i>NLP Project</i> ○ “Extraction of keywords from texts as well as their type” (Python).	Montréal 2021
Polytechnique Montréal <i>Data Visualization Project — Chaire de recherche en fiscalité et en finances publiques</i> ○ “Develop a web application to compare taxes and provincial and federal tax rates by year.” ○ Implement a mockup with Plotly.js, D3.	Montréal 2021
FLEX GROUP <i>Programming Intern</i> ○ “Modeling of Automatic Optical Character Recognition.” ○ Implement models in Python, PHP.	Laval 2019–2020
International Bank of Cameroon for Savings and Credit <i>Intern for the Master's in Applied Statistics</i> ○ “Initiation of mathematical models in forecasting a bank's income and expenses” (R).	Cameroon 2006

Education

Polytechnique Montréal <i>PhD in Computer Engineering</i> ○ AI-driven routing, carbon-aware networking, and zero-trust security for 6G sensor networks.	Montréal, Canada Summer 2026
Polytechnique Montréal <i>Engineering Diploma in Software Engineering</i> ○ Concentration: Artificial Intelligence — Data Sciences.	Montréal, Canada
Advanced National School of Engineering (ENSPY) <i>Master's Degree in Applied Statistics</i> ○ Equivalency: Master's in Probability — Statistics.	Yaoundé, Cameroon

Licenses & Certifications

- AWS Certified Cloud Practitioner (CLF-C01)
Certification Prep Path (Pluralsight), Montréal
- Program Management Professional (PgMP)
Certification Prep Path (Pluralsight), Montréal
- AWS Certified Developer — Associate (DVA-C02)
Certification Prep Path (Pluralsight), Montréal

Conferences & Seminars

PolyCongrès 2026, Polytechnique Montréal

Montréal, Canada

*Oral Presentation**Mar–Apr 2026*

“Energy-Efficient and Near Real-Time Routing for Disaster Monitoring in Wireless Sensor Networks — Graph-Attentive Policy Fusion (GAPF).” Georges Parfait Djimefo, Ranwa Al Mallah, Samuel Pierre.

20th Conf. African Assoc. of Insect Scientists

Yaoundé, Cameroon

*Oral Presentation**Oct 2013*

Djimefo P., Tchuenté M. & Le Gall P. — “Biogeography of communities of insects using clustering methods.” Theme: “Global change and impacts on insect biodiversity: Integrated pest and disease vectors management in Africa.”

29th Annual Conf. Int. Society for Clinical Biostatistics

Copenhagen, Denmark

*Poster**Aug 2008*

“HIV incidence estimation from repeated cross-sectional surveys: a comparative study of parametric and non-parametric approaches by simulation.” Joint work: H. Gwet, R. Pouillot, Y. Le Strat, A. Alioum, J.-C. Thalabard.

ANMSA (African Network of Mathematical Statistics and its Applications)

Franceville, Gabon

*Oral Presentation**Jan 2008*

Université des Sciences et Techniques de Masuku (USTM). “Épidémiologie du VIH au Cameroun: Estimation de l'incidence à partir des données de prévalence.”

Volunteering

- Volunteer at the IEEE branch of Polytechnique Montréal, Montréal, 2018.
- Volunteer at the ConFoo conference, Montréal, 2018.